HERIOT WATT UNIVERSITY
Department of Mathematics

Mock Exam
F18CE2 - Multivariable Calculus and Real Analysis B
Semester 2, 2021/2022

Attempt all questions
Show any calculations you make
(no marks will be given if there is no, or incorrect, working, even if the answer is correct).

1. Are all bounded sequences convergent? Provide a proof or a counterexample.

[2]

State the two conditions that define the infimum.

[2]

Let A be a set of positive real numbers that is bounded above, and consider the set

$$B = \left\{-7a^2 + 1 : a \in A\right\}.$$

Show that B is bounded below and that $\inf B = -7(\sup A)^2 + 1$.

[4]

2. Consider the sequence $\{a_n\}$ given by

$$a_n = \frac{7^n}{(2n+1)!}.$$

Prove that $\{a_n\}$ is decreasing.

[4]

Use this to show that a_n converges, stating fully any theorem you use.

[4]

3. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous at $a \in \mathbb{R}$ and $\lim_{x \to a} f(x) = L > 0$. Prove that there exists a $\delta > 0$ such that $f(x) > 0$ for any $x \in (a - \delta, a + \delta)$.

[5]

4. Use the Intermediate Value Theorem to prove that the equation

$$e^{-x} = x + x^2$$

has a real solution in $[0, 1]$.

[4]

Then use Rolle's theorem to show that it cannot have more than one solution on $\mathbb{R}$.

[5]

5. Use the First Mean Value Theorem to prove that for any integer $n \geq 1$,

$$(n+1)e^n \leq (n+1)e^{n+1} - ne^n \leq (n+2)e^{n+1}.$$

[7]

Note: State clearly any theorems you are using, where you are using the result of these theorems and why they are applicable.

6. Let $\{a_n\}$ be a sequence of real numbers such that $Ar^{-1/2} \leq a_n \leq Br^{-1/2}$ for some constants $A, B > 0$. Discuss the convergence of

(i) $\displaystyle\sum_{r=1}^{\infty} a_r$,

(ii) $\displaystyle\sum_{r=1}^{\infty} a_r^2$,

(iii) $\displaystyle\sum_{r=1}^{\infty} \sqrt{r} a_r^2$,

[6]

stating clearly any theorem you are using.

7. Find the radius of convergence R of the power series

$$\sum_{r=1}^{\infty} \frac{2^r\, x^r}{r^2 + \sqrt{r}}\,.$$

[4]

8. Use the definition of the Riemann integral $\displaystyle\int_0^{1/2} \frac{1}{\sqrt{1-x^2}}\, dx$ to prove that

$$\lim_{n\to\infty} \left(\frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-4}} + \frac{1}{\sqrt{4n^2-9}} + \cdots + \frac{1}{\sqrt{4n^2-n^2}} \right) = \frac{\pi}{6}.$$

[8]

Note: You can assume that $\displaystyle\int_0^{1/2} \frac{1}{\sqrt{1-x^2}}\, dx = \frac{\pi}{6}$ *without computing it.*

9. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined as

$$f(x) = \begin{cases} x^n, & x < 0, \\ x^{n+1}, & x \geq 0. \end{cases}$$

Find the smallest positive integer n for which f is twice differentiable on $\mathbb{R}$.

[7]

Note: You can use the usual rules of calculus to compute the first and second derivatives of the function, without proving their validity.

10. Let (a_n) be a sequence defined by

$$a_n := \left(\frac{1}{2n}\right)^n \qquad \text{for } n \geq 1\,.$$

(i) Find $c > 0$ and $0 < q < 1$ such that

$$a_n \leq cq^{n-1} \qquad \text{for all } n \geq 1\,.$$

(ii) Show with (a) and the geometric progression $\sum_{n=1}^{\infty} cq^{n-1} = \frac{c}{1-q}$ that there exists a constant $U > 0$ such that

$$b_N = \sum_{n=1}^{N} a_n \leq U\,.$$

(iii) Proof with the Monotone Convergence Theorem based on (a) and (b) that the sequence of partial sums (b_N) converges for $N \to \infty$, that means that the series

$$\sum_{n=1}^{\infty} a_n$$

converges.

[8]

End of paper.